

NAPOLEON +

Numerical Assessment of Performance Of next-gen LEO Networks

A User Guide

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Chapter 1

Scenario and Engineering Problem

1.1 Future Mobile Systems & LEO Constellations

Non-Terrestrial Networks (NTNs) are becoming a structural component of future mobile systems. Their role extends beyond emergency connectivity and includes coverage extension, service continuity, network resilience and integration with terrestrial infrastructures. In the 3GPP framework, NTN support is studied within New Radio (NR), considering deployment scenarios, architecture, orbit, propagation conditions, mobility and radio-interface impacts [1]. This perspective defines the domain of NAPOLEON+: LEO satellites are not treated as isolated relays, but as part of an integrated access network. The relevant issue is therefore not only whether satellite coverage exists, but whether users can be dynamically associated to moving satellites while satisfying service-dependent requirements over time.

LEO constellations are especially relevant for broadband and delay-sensitive services because their low altitude reduces propagation delay with respect to geostationary systems. However, the same property makes the access problem highly dynamic. Satellites move rapidly, visibility windows are short, and handovers become frequent. Consequently, LEO access cannot be reduced to a static coverage problem: visibility, elevation angle, slant range, propagation delay and channel quality evolve continuously.

1.1.1 The User-to-Satellite Association Problem

The user-to-satellite association problem consists in selecting, for each user and each simulation instant, the satellite that should serve that user among the visible and admissible candidates. Let $\mathcal{U}$ be the set of users, $\mathcal{S}_u(t)$ the set of satellites available for user u at time t , and $a_u(t)$ the assigned satellite. Conceptually,

$$a_u(t) \in \mathcal{S}_u(t), \quad u \in \mathcal{U}. \quad (1.1)$$

The selected satellite is not necessarily the closest one, nor the one with the highest instantaneous SNR: it depends on the service objective. Broadband traffic may prioritize achievable rate, while latency-sensitive traffic may prioritize delay, link reliability and handover stability. The association problem is therefore dynamic, constrained and service-aware. It must combine physical-layer quantities with system-level requirements, including service continuity and handover cost.

1.1.2 Handover Cost and Service Continuity

Handover is unavoidable in LEO systems because the serving satellite periodically leaves the user's visibility region, but it is costly: each transition may introduce signaling overhead, synchronization procedures, temporary interruption and service degradation. A purely greedy policy that always selects the best instantaneous link can therefore become unstable, as, even when each new link is locally better, excessive switching may reduce service continuity. Such a trade-off is particularly relevant in LEO-NTN access, where uncontrolled handovers may occur every few minutes depending on altitude, minimum elevation angle, user strategy and propagation conditions [2]. Soft-handover studies also highlight that short satellite visibility makes handover management essential for quality of service [3]. For this reason, handover must be treated as the core of the association problem, not as a secondary effect.

1.2 Propagation Environment and Channel-State Uncertainty

The LEO access problem is not determined by orbital geometry alone. The radio link between a ground terminal and a satellite depends on frequency, elevation angle and local propagation environment. Land Mobile Satellite channels are affected by shadowing, blockage, multipath, tropospheric effects and ionospheric effects. ITU-R P.681 provides propagation data for Earth-space land mobile systems, including terrain-induced impairments such as multipath, shadowing and blockage [4]. As a consequence, the satellite with the best geometrical visibility is not necessarily the satellite with the best effective radio link. Low elevation angles increase propagation distance and may worsen blockage and shadowing, while different environments produce different channel statistics.

Another critical issue is Channel-State Information (CSI). Perfect instantaneous CSI is not always realistic in a fast-moving LEO system because feedback may be delayed, incomplete or outdated. At the same time, satellite motion is predictable, so quantities such as visibility time, elevation angle, distance and propagation delay can be estimated in advance. This motivates the comparison between ideal channel knowledge and forecast-based association.

1.3 Service-Aware Access: eMBB and URLLC

Current and future mobile systems are not designed around a single generic service. ITU-R IMT-2020 identifies different usage scenarios, including Enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC) and massive Machine-Type Communications (mMTC) [5]. NAPOLEON+ focuses on eMBB and URLLC because they represent two distinct access objectives.

The eMBB case is throughput-oriented. It represents broadband connectivity, high data-rate services and capacity-driven access. In this case, the relevant indicators are SNR, spectral efficiency and achievable data rate. However, in LEO access, rate maximization must still be balanced against handover stability.

The URLLC case is latency and reliability oriented. Terrestrial URLLC targets cannot be directly transferred to LEO systems because propagation delay is structurally larger than in terrestrial radio access. In NAPOLEON+, URLLC is therefore used as a strict service profile to test whether an association policy can satisfy selected latency, SNR and handover constraints over time.

1.4 Engineering Problem Addressed by NAPOLEON+

The engineering problem addressed by NAPOLEON+ can be summarized as follows:

Given a time-varying LEO constellation, a set of terrestrial users, a propagation model and a service-specific association policy, evaluate whether the resulting user-to-satellite associations remain service-compliant over time.

This formulation captures the three main aspects of the product. First, the problem is dynamic: the simulator must evaluate the temporal evolution of visibility, distance, delay, SNR and rate. Second, the problem is service-aware: the best association depends on whether the target service is throughput-oriented or latency-sensitive. Third, the problem is KPI-driven: the output must be assessed through interpretable indicators. NAPOLEON+ is therefore positioned as a LEO-NTN access evaluation platform for analyzing association policies under mobility, propagation uncertainty and handover constraints.

Chapter 2

NAPOLEON+ Functionalities

2.1 Software Product Scope

To address the engineering problem defined in Sec. 1.4, NAPOLEON+ integrates scenario generation, visibility computation, channel characterization, user-to-satellite association, KPI evaluation and graphical visualization into a single evaluation workflow. The software is conceived as a user-facing access-planning tool rather than as a collection of independent scripts. Its objective is to allow an operator, designer, researcher or system planner to generate a LEO-NTN access scenario, apply a service-aware association policy and evaluate whether the resulting connectivity satisfies the selected performance requirements.

Although the backend is modular, the final product is GUI-driven. The graphical interface acts as the operational layer of the simulator: it collects the high-level input parameters, launches the computational stages, displays the results and exports the generated outputs. The user is therefore not required to interact directly with orbital data structures, channel-model variables or KPI-processing scripts.

2.2 Graphical User Interface as the Access Layer

The GUI is the access layer of NAPOLEON+ and its role is to provide a controlled interface between the user and the 2-stage backend workflow. In the default access-planning mode, the main scenario inputs are limited to:

- the number of simulated terminals;
- the Channel-State Information mode;
- the random seed;
- the default association policy.

This design keeps the interaction at system-planning level. The user can control scenario size, reproducibility, CSI assumption and service objective, while the simulator automatically handles the predefined constellation profile, area of interest, radio configuration and data structures required by the backend. The GUI workflow is organized

around four functional actions: scenario generation, policy tuning, association execution and KPI visualization/export. These actions correspond to the main computational micro-steps of the simulator. The detailed interaction with panels, buttons and modal windows is described in the software user guide in Chapter 3.

2.3 Scenario Generation

Scenario generation is the first computational functionality of NAPOLEON+. Starting from the selected high-level inputs, the simulator builds the dynamic physical and radio-access environment on which the association policy will later operate.

The generated scenario includes:

- the simulation time window and sample time;
- the constellation configuration;
- the area-of-interest configuration;
- the channel configuration;
- the minimum elevation constraint;
- the time-dependent user-satellite visibility map;
- the satellite scenario object used for visualization.

The **default simulation** horizon has a duration of 192 minutes with a sample time of 20 seconds. The random seed ensures reproducibility, which is essential when comparing association policies: the same geometric and channel scenario can be regenerated while changing only the policy configuration.

The **default constellation** profile follows a Starlink-like LEO shell [6], composed of 72 non-polar orbital planes with 22 satellites per plane, inclination 53.2° , phasing parameter 17 and altitude 540 km.

The **default area of interest** covers Europe, with latitude from 35° to 60° and longitude from -10° to 30° . The minimum elevation angle is set to 25° , as enforced by the "Order and Authorization and Order on Reconsideration" FCC-2148 [6].

The output of scenario generation¹ is not a performance result. It is the dynamic link map of the simulated environment, containing the time evolution of visibility, geometry and physical channel quantities (i.e., rate and SNR of every candidate link). Association decisions and KPI computation are performed only in the following stage.

2.3.1 Channel-State Information Modes

NAPOLEON+ implements two Channel-State Information modes: *Ideal* and *Forecast*. They define the type of channel knowledge available to the association algorithm.

¹The scenario, defined as the set of constellation parameters, AoI settings, constraints and default characteristics, is fixed and cannot be tuned from the GUI. These values are nevertheless visible to the user in the configuration table, as described in Subsec. 3.3.1.

Ideal CSI Mode The Ideal CSI mode represents a reference condition in which the association algorithm has access to instantaneous channel information for the visible links. This mode is not intended to represent a fully operational deployment, but to provide a benchmark for the best achievable behavior of a policy under perfect channel awareness.

Forecast CSI Mode The Forecast CSI mode represents a more operationally plausible condition. In this case, the association algorithm relies on averaged channel behavior derived from the expected link geometry given a predetermined set of channel coefficients. This reflects the fact that instantaneous CSI may be delayed, incomplete or outdated in a fast-moving LEO network, while satellite motion, visibility time, slant range and propagation delay remain largely predictable.

The comparison between Ideal and Forecast modes, therefore, allows NAPOLEON+ to separate two effects: the intrinsic performance of the association policy and the artifacts, both in underestimation and overestimation of the link quality, due to imperfect channel knowledge.

2.4 Service-Aware Association Policies

The selected policy determines how candidate satellites are ranked, when a handover is accepted and which service-specific KPIs are evaluated.

2.4.1 From Service Requirements to Association Metrics

The role of the service policy is to translate a communication requirement into a satellite-selection rule. For eMBB, given its service objective, the natural physical-layer indicators are SNR, spectral efficiency and achievable data rate. To control the otherwise excessive handovers that the maximization of instantaneous rate yields, the eMBB association policy balances throughput improvement and service continuity. For URLLC, given that LEO-NTN propagation delay cannot be neglected, and given the relevant service objectives of low latency, reliability, and limited interruption probability, classical terrestrial network URLLC latency targets cannot be transferred directly without qualification. NAPOLEON+ therefore uses URLLC as a latency-sensitive stress-case service profile that enhances, via HO and SNR-awareness, the trivial minimum-distance association, up to appropriate parameter tuning (see Subsec. 2.4.3).

In this sense, for a given optimization problem (in our case, the Munkres algorithm), given the weight associated to the primary metric of the policy (rate and distance, respectively, for eMBB and URLLC) and given the weight associated to the handover event, an HO is accepted only if the candidate satellite provides a sufficient increase in the primary metric² by a certain amount. This quantity is a factor proportional to the input-fixed quantity Δ , which we can think of as a handover margin.

²Further insights and disclosures about the weight policies and the general optimization problem formulation can be found in Subsec. 2.4.4.

2.4.2 eMBB-Oriented Policy

For a candidate link between user u and satellite s , the achievable rate can be expressed as

$$R_{u,s}(t) = B \log_2(1 + \text{SNR}_{u,s}(t)), \quad (2.1)$$

where B is the system bandwidth and $\text{SNR}_{u,s}(t)$ is the link signal-to-noise ratio. The handover margin Δ is such that, if s^* is the best candidate satellite and $a_u(t^-)$ is the current serving satellite, a handover is accepted only if

$$R_{u,s^*}(t) - R_{u,a_u(t^-)}(t) \geq \Delta R_{\text{HO}}, \quad (2.2)$$

where ΔR_{HO} is the minimum rate improvement required to justify the handover. The eMBB policy can thus be tuned by means of the following parameters:

- **Delta rate** ΔR_{HO} : minimum data-rate improvement required to trigger a handover. In its default configuration (end-user does not modify the policy, see Chapter 3) $\Delta R_{\text{HO}} = 2 \text{ Mbit/s}$, that is approximately the 5% of the NTN-NR minimum eMBB compliant bitrate of $R_{\min} = 50 \text{ Mbit/s}$ [1];
- **Minimum rate** $R_{\min}$: target throughput required for service compliance, by default set to the eMBB specs for NTN 5G nets $R_{\min} = 50 \text{ Mbit/s}$;
- **Time window** W : number of samples used to evaluate handover stability, by default equal to 13 samples, or equivalently, for sample time $t_s = 20 \text{ s}$: $W = 13 \cdot 20 \text{ s} \approx 260 \text{ s} \approx 4.4 \text{ min}$. We note that this is approximately the time during which a satellite is visible to a given user, above threshold;
- **Maximum handovers** $H_{\max}$: maximum number of handovers allowed within the selected time window, by default fixed to 4.

2.4.3 URLLC-Oriented Policy

For a candidate link between user u and satellite s , the propagation latency is computed from the slant range $d_{u,s}(t)$ as

$$\tau_{u,s}(t) = \frac{d_{u,s}(t)}{c}, \quad (2.3)$$

where c is the speed of light. Since LEO satellites move rapidly, the satellite with the best instantaneous SNR is not necessarily the satellite with the lowest delay or the most stable service evolution. Thus, an HO is accepted only if the candidate satellite provides a sufficient latency reduction with respect to the current serving satellite:

$$\tau_{u,a_u(t^-)}(t) - \tau_{u,s^*}(t) \geq \Delta \tau_{\text{HO}}, \quad (2.4)$$

where $\Delta \tau_{\text{HO}}$ is the minimum latency improvement required to justify the handover. We note that, in its default configuration, the URLLC policy is more restrictive than the eMBB policy, especially by means of freedom of HOs.

- **Maximum latency** $\tau_{\max}$: maximum tolerated service latency, by default is fixed to $\tau_{\max} = 3 \times 10^{-3} \text{ s}$. This threshold is to intend as a propagation-delay proxy rather than as a full end-to-end URLLC requirement. Since $\tau = d/c$, $\tau_{\max}$ corresponds

to a maximum acceptable slant range of approximately 900 km. In the considered 540 km LEO scenario, visible (above threshold) links span roughly from 1.8 ms to 4.26 ms; therefore, the selected latency threshold admits close and medium-elevation satellites while rejecting low-elevation links with larger propagation delay;

- **Delta latency** $\Delta\tau_{\text{HO}}$: minimum latency reduction required to trigger a handover, by default set to $\Delta\tau_{\text{HO}} = 1 \times 10^{-3}s$, so to enable meaningful HOs and prevent both ping-pong effect and coarse resilience to convenient HOs;
- **Minimum SNR** γ_{min} : minimum link-quality threshold for service compliance, by default fixed to $\gamma_{\text{min}} = 10$ as a conservative robustness threshold, preventing the association from considering links that are latency-compliant but too weak in terms of received signal quality, thus unreliable by means of error rates;
- **Time window** W : number of samples used to evaluate handover stability, by default fixed to approximately half a satellite pass, that is $W = 6 : t = t_s W = 120s$;
- **Maximum handovers** H_{max} : maximum number of handovers allowed within the selected time window, by default fixed to 1 to measure reliability by penalizing frequent (even if convenient) HOs.

2.4.4 Association Controls and KPI Controls

NAPOLEON+ separates association-control parameters from KPI-control parameters. Association-control parameters influence the decisions made during the simulation, that is they determine the weight of each policy in the optimization matrix built to exploit the Hungarian algorithm. Namely, those parameters are ΔR_{HO} for eMBB and $\Delta\tau_{\text{HO}}$ for URLLC.

As the product NAPOLEON+ is built for highly aware end-users, that is, domain experts and professionals, it is of relevance to share the expression of said weights, so to enable operators of every such kind to test new policies by consciously varying Δ . Thence, for the optimization problem:

$$\min_{\mathbf{x}^{(t)}} \sum_{u=1}^U \sum_{s=1}^S M_{u,s}^{(t)} x_{u,s}^{(t)},$$

the entry $M_{u,s}^{(t)}$ of the cost matrix is defined as

$$w_d D_{u,s}^{(t)} + w_R Q_{u,s}^{(t)} + w_{\text{HO}} H_{u,s}^{(t)}, \quad w_d + w_R + w_{\text{HO}} = 1,$$

where, for eMBB:

$$w_R = \frac{R_{\text{max}}}{R_{\text{max}} + \Delta R_{\text{switch}}}, \quad w_{\text{HO}} = \frac{\Delta R_{\text{switch}}}{R_{\text{max}} + \Delta R_{\text{switch}}}, \quad w_d = 0,$$

and, for URLLC:

$$w_d = \frac{d_{\text{max}} - d_{\text{min}}}{d_{\text{max}} - d_{\text{min}} + c\Delta\tau_{\text{switch}}}, \quad w_{\text{HO}} = \frac{c\Delta\tau_{\text{switch}}}{d_{\text{max}} - d_{\text{min}} + c\Delta\tau_{\text{switch}}}, \quad w_R = 0.$$

KPI-control parameters on the other hand are used after the association has been computed to evaluate whether the obtained service satisfies the selected requirements: (i) $R_{\min}, H_{\max}$ for eMBB, and (ii) $\tau_{\max}, \gamma_{\min}, H_{\max}$ for URLLC. We must reinforce the notion that these thresholds **do not change the selected satellite** at each instant; instead, they define the compliance conditions used in the performance assessment, thus preventing the tuning interface from mixing decision logic and evaluation logic.

2.5 Association Execution Engine

After scenario generation, the GUI enables the association stage. The execution engine receives the generated scenario and the current parameter structure, then runs the selected policy and computes the corresponding KPIs.

The association stage performs three operations:

1. it builds the association configuration from the selected policy and radio parameters;
2. it runs the user-to-satellite association module;
3. it computes the general and service-specific KPIs.

The association configuration includes the selected policy, carrier frequency, transmit power, antenna gains, bandwidth, noise power, minimum elevation angle, satellite altitude, and capacity constraint³. The KPI configuration contains the thresholds associated with the selected service class.

The result structure stores the complete association history, including assigned satellite index, SNR, rate, distance, latency, handover events and served-user mask. These outputs are then converted into plots, tables and compliance indicators by the GUI.

2.6 KPI Computation and Visualization

KPI computation is the final analytical layer of NAPOLEON+, and it is part of the second stage. Its purpose is to transform raw association outputs into interpretable performance indicators. The simulator computes both general KPIs, common to all policies, and service-specific KPIs, activated according to the selected policy.

The general KPI set includes:

- SNR evolution averaged over the users;
- throughput evolution averaged over the users;
- cumulative distribution of average user rate (rate fairness);
- aggregate handover evolution;
- served-user fraction over time;

These quantities describe the global behavior of the association process. They allow the user to assess link quality, broadband performance, handover intensity and service availability over the simulation horizon.

³The radio and architecture parameters purpose is the computation of Min-Max normalization variables on the objective's quantities, so to ensure the absence of off-scale biases.

2.6.1 Service Compliance

Service-specific KPIs evaluate whether the association output satisfies the active service requirements.

For eMBB, a user u is compliant at time t if its assigned rate is above the required threshold and the number of handovers in the evaluation window does not exceed the allowed maximum:

$$C_{\text{eMBB}}(u, t) = \mathbf{1}[R_u(t) \geq R_{\min}] \mathbf{1}[H_u(t; W) \leq H_{\max}], \quad (2.5)$$

where $R_u(t)$ is the achieved rate of user u , $H_u(t; W)$ is the number of handovers performed by user u within the window W , and $\mathbf{1}[\cdot]$ is the indicator function.

In addition to compliance, eMBB performance is also evaluated through the aggregate system spectral efficiency. Let B be the system bandwidth and let $\tilde{R}_u(t)$ denote the effective delivered rate, set to zero if the user is not served or if the rate value is not finite. The system spectral efficiency is defined as

$$\frac{1}{B} \sum_{u \in \mathcal{U}} \tilde{R}_u(t), \quad (2.6)$$

and is expressed in bit/s/Hz, that is the aggregate throughput delivered by the association strategy per unit bandwidth.

Similarly for URLLC, recalling its objective, the compliance reads as:

$$C_{\text{URLLC}}(u, t) = \mathbf{1}[\tau_u(t) \leq \tau_{\max}] \mathbf{1}[\gamma_u(t) \geq \gamma_{\min}] \mathbf{1}[H_u(t; W) \leq H_{\max}], \quad (2.7)$$

where $\tau_u(t)$ is the propagation latency and $\gamma_u(t)$ is the received SNR of the assigned link. The temporal compliance ratio is then computed as the fraction of users satisfying the service condition at each time instant:

$$\text{TCR}_{\text{service}}(t) = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} C_{\text{service}}(u, t). \quad (2.8)$$

Since average latency alone can hide poor tail behavior, URLLC is also evaluated through a percentile latency metric. For each time instant t , the p -th percentile latency is computed over the set of served users with finite latency values:

$$Q_p(\{\tau_u(t) : u \in \mathcal{U}, \text{served}_u(t) = 1\}), \quad (2.9)$$

where $Q_p(\cdot)$ denotes the p -quantile operator. In the implemented configuration, $p = 0.90$, so $P_{\tau, \text{URLLC}}(t)$ represents the latency below which 90% of the served users lie at time t .

A global percentile latency is also computed over all served user-time latency samples:

$$Q_p(\{\tau_u(t) : u \in \mathcal{U}, t \in \mathcal{T}, \text{served}_u(t) = 1\}), \quad (2.10)$$

thus, acting as an aggregate parameter, displayed to the user accordingly to Fig. 2.1b.

Mean eMBB TCR	-	%

(a) eMBB summary table content.

URLLC 90th Percentile	-	ms
Mean URLLC TCR	-	%

(b) URLLC summary table content.

Figure 2.1: Service-dependent portion of the KPI Summary table.

2.6.2 KPI Summary and User-Level Inspection

In addition to graphical plots, NAPOLEON+ provides tabular KPI summaries. The summary table reports aggregate (time averaged) quantities over the full simulation, both by means of service-specific KPIs (see Fig. 2.1) and general KPIs (see Fig. 2.2), giving an immediate overview of the result.

System-level KPIs however may hide local service problems. A policy can show good average performance while specific users experience outage, low throughput or excessive handovers. To address this issue, NAPOLEON+ includes a user-level inspection functionality. The User Inspector allows the operator to select a terminal by user ID and visualize two easy-to-interpret and high value user-specific metrics: (i) available rate evolution overtime and (ii) service-state evolution, which highlights the cumulative distribution of handovers overtime, the total amount of HOs, the overall service percentage and the time instants when experiencing lack of service. This supports debugging, policy interpretation and local service analysis.

Metric	Value	Unit
Avg User SNR	-	dB
Avg User Rate	-	Mbps
Avg User Spectral Efficiency	-	b/s/Hz
Mean Total Handovers	-	HO/step

Figure 2.2: KPI summary table portion for general KPIs

2.7 Export and Reproducibility

NAPOLEON+ includes an export functionality activated after the association is completed. The export procedure saves both processed and raw information.

The exported package includes:

- an Excel report containing the configuration and KPI summary tables;
- a MATLAB raw-data file containing the results, scenario and parameter structures;
- high-resolution PNG figures for the main KPI plots;
- the user-distribution map.

The raw-data export is important for reproducibility. It allows the generated scenario and association results to be stored without rerunning the simulation. Together with the random seed, this makes NAPOLEON+ suitable for controlled comparison between different policies, CSI assumptions or parameter settings.

Chapter 3

Software User Guide

3.1 Launching NAPOLEON+

The NAPOLEON+ simulator can be launched either from MATLAB by running the `NAPOLEON_GUI_v0.m` script, or through the compiled standalone application¹. The following content describes how an end-user operates the interface. The modeling assumptions, association logic, and KPI definitions are presented in Chapter 2.

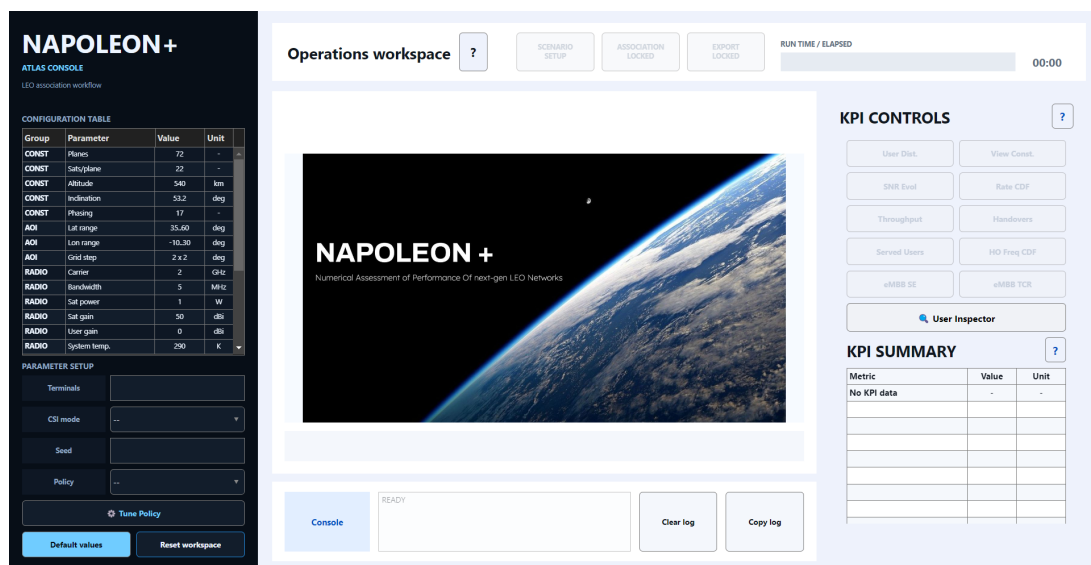


Figure 3.1: NAPOLEON+ main window. The interface is divided into configuration controls, visualization workspace and analytics panel.

3.2 Interface Layout

As shown in Figure 3.1, the GUI is organized into three working areas:

- **Left panel:** scenario inputs, policy selection, policy tuning and configuration table;
- **Center panel:** main visualization area for maps, plots and execution feedback;

¹More about the standalone application and its execution in Sec. ??

- **Right panel:** KPI buttons, user inspection and summary table.

The interface follows the execution order of the simulator: configure the scenario, generate it, run the association, inspect the KPIs and export the results.

3.3 Input Configuration

The input configuration phase defines the scenario parameters and the active association policy before the computational run.

3.3.1 Configuration Table

The configuration table, located in the left panel, summarizes the parameters currently used by the simulator. It is organized into fixed and tunable groups.

- **CONST:** constellation parameters;
- **AOI:** area-of-interest parameters;
- **RADIO:** radio and link-budget parameters;
- **RUN:** user-selected scenario inputs;
- **eMBB / URLLC:** active policy thresholds.

The table is used as a compact check of the active configuration before running the association stage.

3.3.2 Scenario Parameters

The *Parameter Setup* box allows the user to set the main scenario inputs:

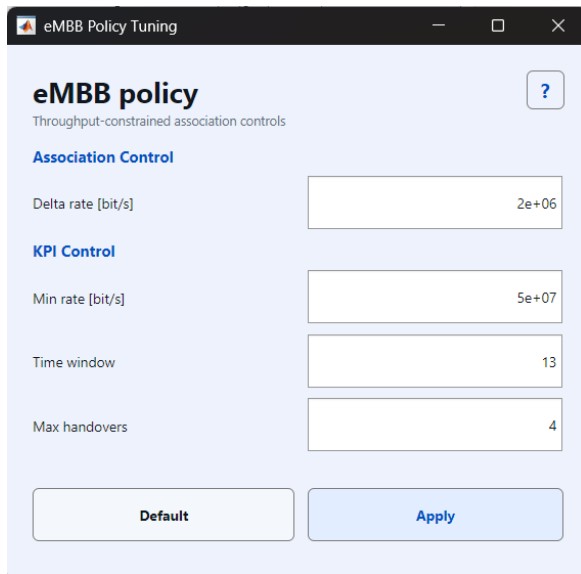
- **Terminals:** number of terrestrial users generated in the area of interest;
- **CSI Mode:** channel-state used by the association stage (*Ideal* or *Forecast*);
- **Random Seed:** seed used to reproduce the same user distribution and scenario realization.

After these fields are set, the scenario can be generated through the execution controls described in Section 3.4.

3.3.3 Association Policy

The **Policy** dropdown selects the active association policy:

- **eMBB;**
- **URLLC.**



eMBB policy ?

Throughput-constrained association controls

Association Control

Delta rate [bit/s]

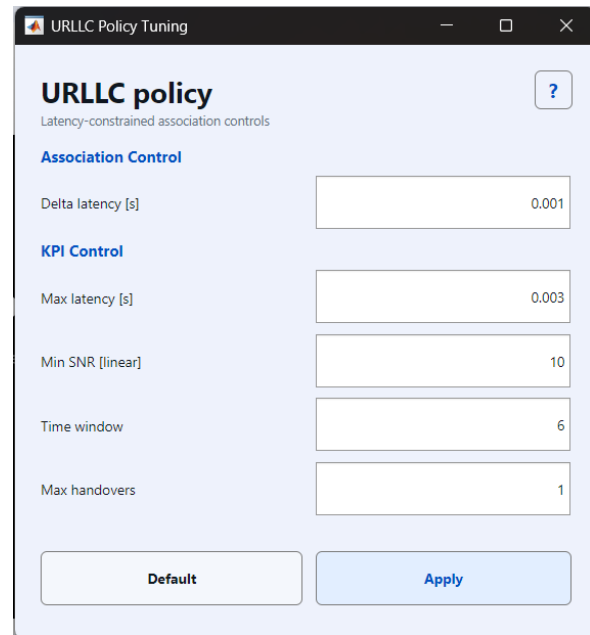
KPI Control

Min rate [bit/s]

Time window

Max handovers

(a) eMBB tuning window.



URLLC Policy Tuning ?

URLLC policy

Latency-constrained association controls

Association Control

Delta latency [s]

KPI Control

Max latency [s]

Min SNR [linear]

Time window

Max handovers

(b) URLLC tuning window.

Figure 3.2: Policy tuning windows.

The **Tune Policy** button opens the policy-specific tuning window. The available controls depend on the selected policy.

For the eMBB policy, the tunable parameters are:

- **Delta rate:** minimum rate improvement required for a handover;
- **Minimum rate:** threshold used for eMBB compliance evaluation;
- **Time window:** number of samples used for handover-window evaluation;
- **Maximum handovers:** maximum allowed handovers inside the selected window.

For the URLLC policy, the tunable parameters are:

- **Delta latency:** minimum latency reduction required for a handover;
- **Maximum latency:** latency threshold used for URLLC compliance evaluation;
- **Minimum SNR:** minimum SNR threshold used for URLLC compliance evaluation;
- **Time window:** number of samples used for handover-window evaluation;
- **Maximum handovers:** maximum allowed handovers inside the selected window.

3.4 Operations Workspace and Execution

The operations workspace controls the simulation sequence.

The available actions are:

- **SCENARIO RUN:** generates the scenario from the selected inputs;

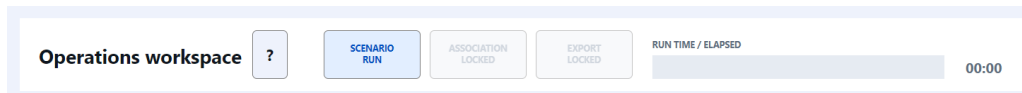


Figure 3.3: Operations workspace.

- **ASSOCIATION RUN:** runs the selected association policy on the generated scenario;
- **EXPORT:** saves the simulation outputs after the association stage;
- **RUN TIME progress bar:** displays execution progress and elapsed time.

The workflow is sequential. The association run becomes available only after a successful scenario generation. The export command becomes available only after the association has been completed. Policy parameters can be modified after scenario generation; in that case, the same scenario can be reused and only the association stage must be run again.

3.5 Performance Analytics

After the association run, the right panel activates the KPI controls and updates the summary table.

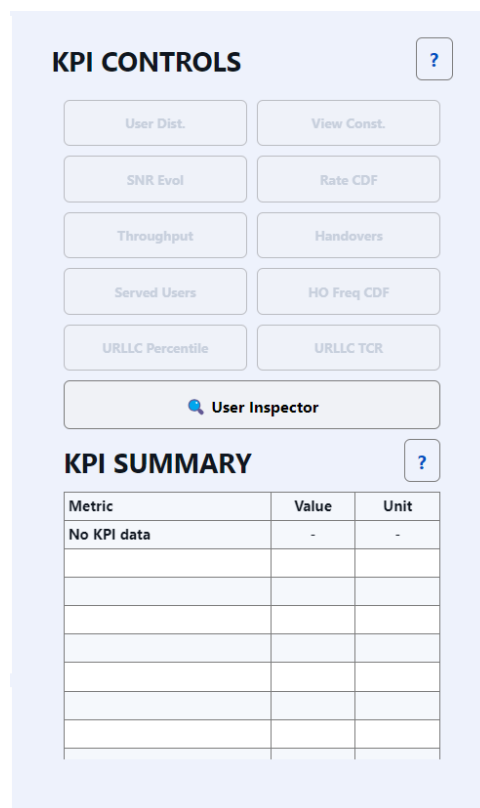


Figure 3.4: KPI controls and summary table.

3.5.1 KPI Controls

The KPI buttons display the selected output in the central workspace or in a dedicated visualization window.

Table 3.1: KPI control buttons.

Button	Displayed output
User Dist.	Geographic distribution of terrestrial users
View Const.	3D constellation visualization
SNR Evol.	Average SNR evolution over time
Throughput	Average throughput evolution over time
Rate CDF	CDF of average user rate
HO Freq CDF	CDF of per-user handover frequency
Handovers	Number of handover events over time
Served Users	Served-user fraction over time

The two bottom KPI buttons depend on the active policy.

Table 3.2: Policy-specific KPI controls.

Policy	Button	Displayed output
URLLC	URLLC Percentile	Latency 90-th percentile over time
URLLC	URLLC TCR	URLLC temporal compliance ratio
eMBB	eMBB Spec. Eff.	System spectral efficiency over time
eMBB	eMBB TCR	eMBB temporal compliance ratio

The compliance definitions used by the TCR plots are given in Section 2.6.1.

3.5.2 User Inspector Tool

The **User Inspector** provides single-terminal inspection. The user selects a terminal ID, and the GUI generates user-specific plots.

The available outputs are:

- **Rate evolution:** instantaneous rate assigned to the selected user over time;
- **Service-state evolution:** connection state and handover timeline for the selected user.

This tool is useful when a global KPI suggests a degradation and the user needs to inspect individual terminal behavior.

3.5.3 KPI Summary Table

The KPI summary table reports aggregate values over the completed simulation. The standard entries are:

- **Avg User SNR;**

- Avg User Rate;
- Avg Spectral Efficiency;
- Mean Total Handovers.

Additional entries depend on the selected policy:

- **eMBB**: Mean eMBB TCR;
- **URLLC**: URLLC latency percentile and Mean URLLC TCR.

The numerical interpretation of these quantities in a complete simulation campaign is discussed in Chapter 4.

3.6 Export

Once the association stage is complete, the **EXPORT** command saves the available outputs for external analysis and reproducibility. The exported content is described in Section 2.7.

3.7 Software Installation and Usage

The NAPOLEON+ Simulator is hosted on GitHub and can be executed either as a standalone desktop application or directly from its MATLAB source code, requesting access from the [repository owner](#).

Running as a Standalone Application For users without a MATLAB license, the simulator has been compiled into standalone desktop applications for both Windows and macOS platforms.

1. Navigate to the *Releases* section of the official GitHub repository.
2. Download the appropriate setup file for your Operating System:
 - **Windows**: Download the Windows installer (e.g., `Napoleon_installer_win.exe`).
 - **macOS**: Download the Mac archive (e.g., `Napoleon_installer_MacOs.zip`) and extract the application bundle.
3. Run the installer (or launch the extracted application on Mac). If your system does not already have the necessary MATLAB Runtime installed, the setup wizard will automatically download and configure it for free over the internet.
4. Once the setup is complete, launch the **NAPOLEON+ Simulator** directly from your desktop or Applications folder.

Running from Source Code For researchers intending to modify the codebase or inspect the underlying orbital mechanics and optimization algorithms, the simulator can be executed directly via the MATLAB environment. It must be noted that NAPOLEON+ requires MATLAB R2024a or later and the Satellite Communications Toolbox.

1. Clone or download the repository to your local machine.
2. Open MATLAB and navigate to the project's root directory.
3. Double-click the `NAPOLEON_plus_1.prj` project file. This action will automatically configure the MATLAB path, load the *resources* folder, and link all necessary dependencies seamlessly across any operating system.
4. Execute the `NAPOLEON_GUI_v0.m` script located in the `src/GUI` directory to launch the interactive canvas.

Chapter 4

Simulation Use Case Study

4.1 NAPOLEON+ Investigation Modes

One of the core design philosophies behind the NAPOLEON+ framework is its versatility. The simulator is engineered to address the core challenge of service-aware connectivity in Non-Terrestrial Networks from multiple professional perspectives:

- **Access Planner mode:** Tailored for *Network Operators* who need to answer topological questions, such as whether the current constellation is physically capable of providing service-aware connectivity over a specific geographic region (e.g., Europe) for eMBB or URLLC users.
- **Compliance Dashboard mode:** Meant for *Service Providers* and focused strictly on Quality of Service (QoS). This mode is used to monitor the number of users, in various scenarios, who successfully satisfy the strict service level constraints over time.
- **Policy Lab and Design mode:** Designed for *Algorithm Designers*. It allows engineers to tweak the mathematical weights and logic inside the backend to define entirely new association policies, observing how optimization shifts impact the network.
- **LEO Explorer mode:** Intended for *Researchers*. This mode investigates the profound fundamental questions of NTN communications: How do LEO orbital dynamics, Channel State Information (CSI), handovers, and association margins directly affect overall network performance?

This chapter focuses exclusively on a **LEO Explorer mode** use case, so to demonstrate the software's capability to isolate and analyze the impact of real-world physical constraints on the performance of associations algorithms.

4.2 Simulation Campaign Setup

In this specific campaign, our goal is to evaluate the impact of channel knowledge (CSI) on the association algorithms.

For the purpose of this example, we focus only on the eMBB-based association policy, but the same approach could be followed also with the URLLC-based policy.

The investigation follows a strict comparative methodology:

1. **Scenario Freezing:** To perform a fair differential analysis, the physical environment must be completely deterministic. We fix the total number of terrestrial users (e.g., $N = 100$) and the Random Generation Seed (e.g., $\text{Seed} = 13$). By doing so, the geographic distribution of the terminals and the exact orbital trajectories of the satellites remain identical across all subsequent runs.
2. **Theoretical Upper-Bound (Ideal CSI):** The selected association policy (eMBB) is first executed under the *Ideal CSI* mode. This initial run establishes the theoretical maximum performance of the network, serving as our perfect baseline.
3. **Real-World Impact (Forecast CSI):** Without changing the parameters fixed before, the CSI mode is simply switched to *Forecast*, and the simulation is re-run. The algorithm is now forced to make handover decisions based on foreseen channel state informations, accurately simulating the physical constraints of a real Non-Terrestrial Network.

4.3 Results Interpretation and CSI mode Comparison

To rigorously evaluate the impact of channel knowledge on the association algorithm, this analysis focuses on a specific set of Key Performance Indicators: Average User SNR, Average User Throughput, Rate CDF (Fairness), the eMBB Temporal Compliance Ratio (TCR), and Handover Frequency. These metrics have been extracted and plotted for both the Ideal and Forecast CSI modes to provide a direct differential visual comparison.

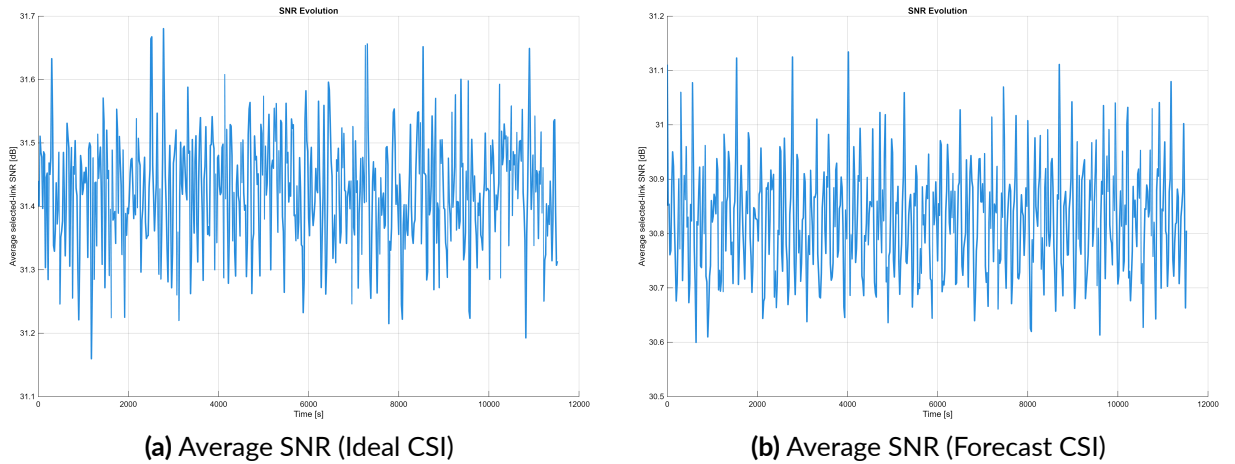
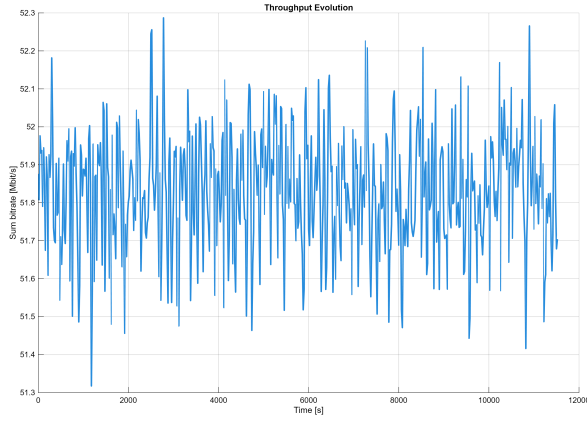
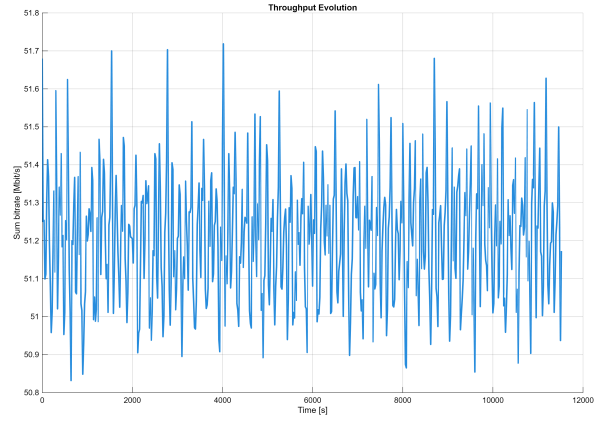


Figure 4.1: Temporal evolution of the Average User SNR. As expected, the Ideal mode achieves a slightly higher global mean (31.42 dB vs. 30.83 dB) but exhibits significantly larger temporal fluctuations, as the algorithm aggressively tracks instantaneous channel peaks.

As illustrated in Figure 4.1, the Average SNR demonstrates how the Ideal CSI algorithm aggressively tracks instantaneous channel peaks, resulting in a higher and more unstable signal quality. Let us now observe the resulting Average User Throughput.



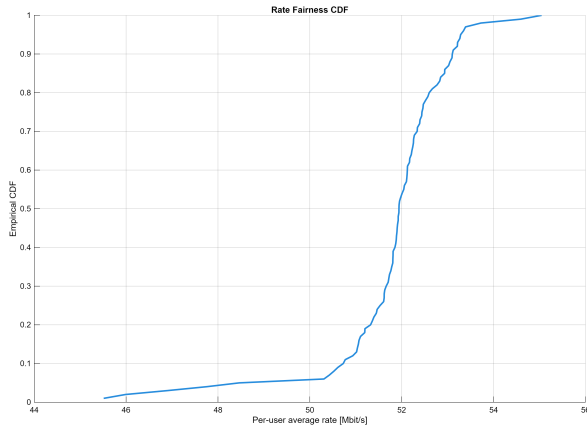
(a) Average Throughput (Ideal CSI)



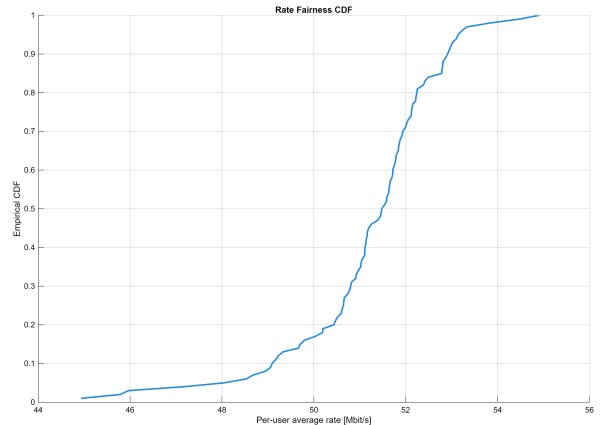
(b) Average Throughput (Forecast CSI)

Figure 4.2: Temporal evolution of the Average User Throughput. As a direct consequence of the SNR behavior, the Ideal mode delivers a slightly higher global mean (51.82 Mbps) compared to the Forecast mode (51.20 Mbps), while inheriting the same highly oscillatory pattern.

While the temporal graphs suggest a similar macroscopic throughput performance between the two modes, global averages can hide underlying disparities. To evaluate whether this capacity is fairly distributed across the entire terminal population, we must examine the Rate Cumulative Distribution Function (CDF).



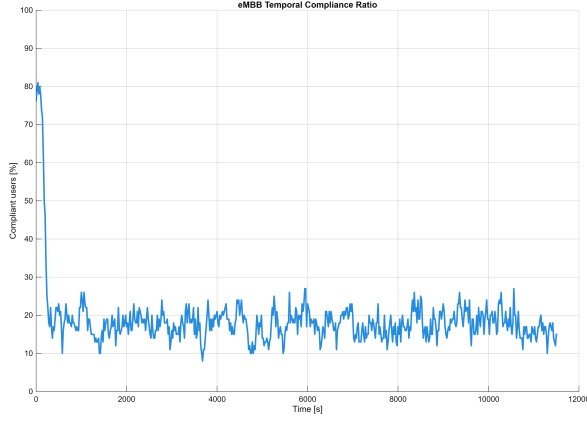
(a) Rate CDF (Ideal CSI)



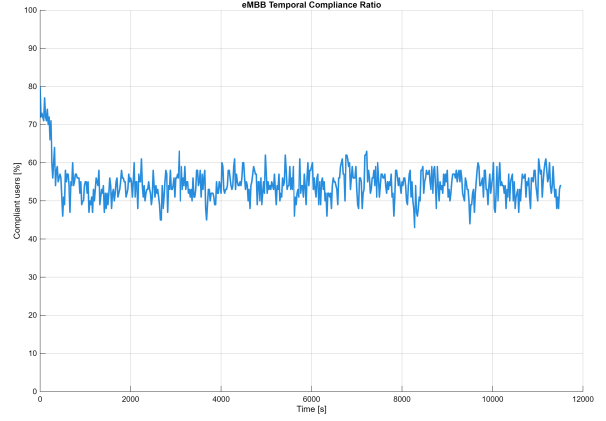
(b) Rate CDF (Forecast CSI)

Figure 4.3: Cumulative Distribution Function of the per-user average rates. Both CSI modes exhibit a nearly identical slope, indicating a comparable level of distributional fairness across the terminal population. However, the Ideal mode (left) curve is shifted to the right, directly reflecting its ability to guarantee a slightly higher data rate on average thanks to perfect instantaneous channel knowledge.

As expected, the Ideal CSI establishes the theoretical upper bound for physical capacity of the network. However, true Quality of Service in Non-Terrestrial Networks is not exclusively about maximizing raw data rates; it fundamentally requires service continuity. To evaluate how CSI mode affects the delivered service, we must shift our focus to the Temporal Compliance Ratio (TCR).



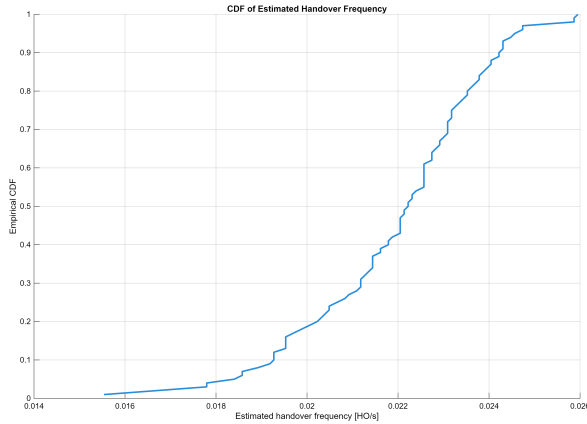
(a) eMBB TCR (Ideal CSI)



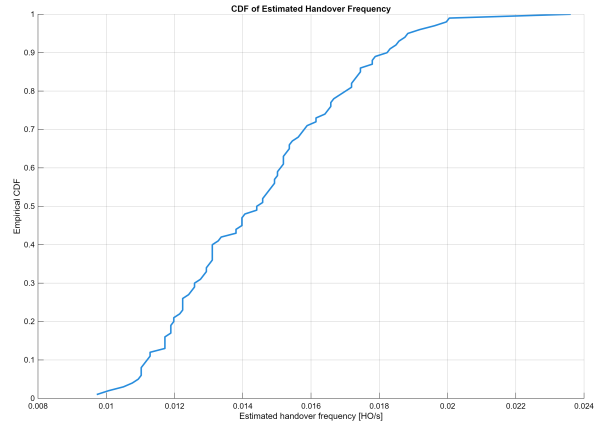
(b) eMBB TCR (Forecast CSI)

Figure 4.4: Temporal evolution of the eMBB Temporal Compliance Ratio. The Ideal mode (left) experiences a severe collapse in overall TCR (18.71%) compared to the much more stable Forecast mode (54.49%).

As evident from the previous plots, the theoretical superiority of the Ideal CSI does not translate into a reliable service, resulting instead in a catastrophic drop in QoS compliance. Since the Rate CDF demonstrated that both modes deliver sufficient and fairly distributed throughput, this collapse must be exclusively caused by the violation of the other constraint: the maximum handover limit ($HO_{\max}$). To verify this hypothesis, we now inspect the Handover Frequency CDF.



(a) Handover Frequency CDF (Ideal CSI)



(b) Handover Frequency CDF (Forecast CSI)

Figure 4.5: Cumulative Distribution Function of Handover Frequencies. The Ideal mode (left) curve is visibly shifted towards higher frequencies (up to 0.026 HO/s), visually confirming the massive explosion in handover events compared to the Forecast mode (right).

The severe TCR collapse under Ideal CSI is strictly caused by the violation of the maximum handover threshold. In Figure 4.5 the x-axis represents the global average handover rate per user in HO/s. The Ideal mode curve is visibly shifted towards higher frequencies, with a median around ≈ 0.022 HO/s: this means an average user is forced to perform a satellite switch roughly every 45 seconds ($1/0.022$). In contrast, the Forecast mode exhibits a median frequency of approximately 0.0145 HO/s (one handover every 66 seconds), providing far greater stability. Using the default values for the computation of TCR, a user is compliant if its handover frequency does not exceed ≈ 0.015 HO/s. This explain why in the Ideal case the TCR has such a low values.

The physical reason behind this excessive switching rate in Ideal mode lies in the temporal resolution of the channel knowledge. Possessing perfect real-time visibility of instantaneous fading, the Ideal algorithm becomes hyper-reactive: it constantly triggers "ping-pong" handovers to exploit short-lived signal spikes. This greediness slightly improves raw throughput but ultimately destroys service continuity. Conversely, Forecast estimation acts as a mathematical low-pass filter, "blurring" out high-frequency fluctuations. This prevents temporary signal spikes from triggering unnecessary handovers, anchoring users to their current satellites and ensuring high QoS compliance.

Iterative Optimization: The Role of Δ Rate To mitigate the ping-pong anomaly, a network engineer can tune the Δ Rate margin. A higher Δ Rate demands a massive capacity gain to justify a satellite switch, effectively forcing the algorithm to ignore micro-fluctuations. This intentionally sacrifices absolute peak throughput in favor of stability, drastically suppressing the handover count and recovering the global TCR.

4.4 A Final Note on: Use Case and Simulation Campaign(s)

We shall conclude by noting that the interpretation of results provided in the previous section, as briefly anticipated, has been conducted assuming that the researcher (the end-user in the "LEO Explorer mode" scenario) would focus in the analysis of SNR, rate, and compliance, and extending its understanding of the system behavior by evaluating the Rate fairness and HO frequency distribution.

This choice does not limit the actual interaction possibilities offered to the end-user, as it is a modeling and narrative strategy to describe possible clients and has no real translation nor limitations by means of UI/UX. The GUI allows in fact the user to inspect any of the KPIs introduced in Sec. 2.6, through the dedicated analytics panel described in Sec. 3.5.

Therefore, the adopted narrative should be understood as one representative interpretation of the simulation campaign: it reflects the natural priorities of an academic/research-oriented use case, without restricting the generality of the software workflow.

For completeness, we recall the four use cases introduced in Sec. 4.1 and focus on the remaining three: (i) the **Access Planner** investigation mode, (ii) the **Compliance Dashboard** investigation mode, and (iii) the **Policy Lab and Design** investigation mode.

For each mode, the engineering question defining the use case naturally leads to the following simulation campaigns:

- **Access Planner:** 2-step simulation campaign, one for each (default) policy. The network operator fixes the number of users N, N' expected to require a given service over the AoI, sets the seed, and runs one simulation per policy.
- **Compliance Dashboard:** M -step simulation campaign, one for each user population size. The service-provider employee selects a set of M possible values of N , for a fixed policy and seed.
- **Policy Lab and Design:** X -step simulation campaign, one for each policy configuration. The algorithm designer selects a policy and a set of X configurations, each defined by its Δ parameter and KPI-control tuple.

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- [3] H. Ben Salem, A. Tarable, A. Nordio, and B. Makki, “Uplink Soft Handover for LEO Constellations: How Strong the Inter-Satellite Link Should Be,” 2025.
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- [6] Federal Communications Commission, “Space Exploration Holdings, LLC: Request for Modification of the Authorization for the SpaceX NGSO Satellite System,” Federal Communications Commission, Order and Authorization and Order on Reconsideration FCC 21-48, Apr. 2021, IBFS File No. SAT-MOD-20200417-00037; Call Signs S2983 and S3018; Adopted April 23, 2021; Released April 27, 2021.